

Mathematics

Presentation

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Diagonalization of a Matrix Theorem

Theorem : If a square matrix A of order n has n linearly independent eigen vectors , then a matrix 'P' can be found such that $P^{-1}AP$ is a diagonal matrix.

PROOF : Let A is the matrix.

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

And let $\lambda_1, \lambda_2, \lambda_3$ be its eigen values and x_1, x_2, x_3 the corresponding eigen vectors where

$$x_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} \quad x_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} \quad x_3 = \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix}$$

Characteristic equation of A is $|A - \lambda I| = \begin{vmatrix} a_1 - \lambda & b_1 & c_1 \\ a_2 & b_2 - \lambda & c_2 \\ a_3 & b_3 & c_3 - \lambda \end{vmatrix}$



$$\begin{bmatrix} (a_1 - \lambda_1) & b_1 & c_1 \\ a_2 & (b_2 - \lambda_1) & c_2 \\ a_3 & b_3 & (c_3 - \lambda_1) \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = 0$$

$$(a_1 - \lambda_1)x_1 + b_1 y_1 + c_1 z_1 = 0$$

$$a_2 x_1 + (b_2 - \lambda_1) y_1 + c_2 z_1 = 0$$

$$a_3 x_1 + b_3 y_1 + (c_3 - \lambda_1) z_1 = 0$$

We have ,

$$a_1 x_1 + b_1 y_1 + c_1 z_1 = \lambda_1 x_1$$

$$a_2 x_1 + b_2 y_1 + c_2 z_1 = \lambda_1 y_1 \quad \dots\dots\dots 1$$

$$a_3 x_1 + b_3 y_1 + c_3 z_1 = \lambda_1 z_1$$

Similarly, for λ_1 and λ_2 ,we have

$$a_1 x_2 + b_1 y_2 + c_1 z_2 = \lambda_2 x_2$$

$$a_2 x_2 + b_2 y_2 + c_2 z_2 = \lambda_2 y_2 \quad \dots\dots\dots 2$$

$$a_3 x_2 + b_3 y_2 + c_3 z_2 = \lambda_2 z_2$$

$$a_1 x_3 + b_1 y_3 + c_1 z_3 = \lambda_3 x_3$$

$$a_2 x_3 + b_2 y_3 + c_2 z_3 = \lambda_3 y_3 \quad \dots\dots\dots 3$$

$$a_3 x_3 + b_3 y_3 + c_3 z_3 = \lambda_3 z_3$$



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We consider the matrix $P = \begin{bmatrix} x1 & x2 & x3 \\ y1 & y2 & y3 \\ z1 & z2 & z3 \end{bmatrix}$

, whose columns are the eigen vectors of A.

• Then, $AP = \begin{bmatrix} a1 & b1 & c1 \\ a2 & b2 & c2 \\ a3 & b3 & c3 \end{bmatrix} \begin{bmatrix} x1 & x2 & x3 \\ y1 & y2 & y3 \\ z1 & z2 & z3 \end{bmatrix}$

From solving this we get ,

$$\begin{bmatrix} \lambda_1 x1 & \lambda_2 x2 & \lambda_3 x3 \\ \lambda_1 y1 & \lambda_2 y2 & \lambda_3 y3 \\ \lambda_1 z1 & \lambda_2 z2 & \lambda_3 z3 \end{bmatrix} \quad \text{Using eq}^n \quad 1, 2, 3$$

$$\begin{bmatrix} x1 & x2 & x3 \\ y1 & y2 & y3 \\ z1 & z2 & z3 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} = PD \quad (\text{where D is a diagonal matrix})$$

Now , finally

$$AP = PD$$

$$\Rightarrow P^{-1}AP = P^{-1}PD = D$$

Hence , it is proved



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Let's take example to understand this more,

Ex : verify $P^{-1}AP = D$, $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ where D is the diagonal matrix.

- solution : Characteristic equation $|A - \lambda I| = 0$

$$\begin{vmatrix} 4 - \lambda & 1 \\ 2 & 3 - \lambda \end{vmatrix} = 0$$

After solving we get, $\lambda = 2, \lambda = 5$

Eigen values are 2 and 5.

(i) When $\lambda = 2$, eigen vectors are

$$\begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

$$2x_1 + x_2 = 0 \Rightarrow x_2 = -2x_1 \quad (x_1 = k, x_2 = -2k)$$

$$x_1 = \begin{bmatrix} k \\ -2k \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

(ii) When $\lambda = 5$, eigen vectors are

$$\begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-x_1 + x_2 = 0 \Rightarrow x_1 = x_2 \quad (x_1 = k, x_2 = k)$$

so, $x_2 = \begin{bmatrix} 1 \\ k \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$



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Model matrix $P = \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix}$

now , we know that $P^{-1} = \text{adj } P / |P|$

- $\Rightarrow P^{-1} = 1/3 \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$

For diagonalization ,

$$D = P^{-1}AP$$

$$\Rightarrow 1/3 \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$$



Thank you



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